

COMPUTER SCIENCE TRIPOS Part IB – 2021 – Paper 6

6 Computation Theory (amp12)

Lambda Calculus A set  $A$  equipped with a binary operation  $@ : A \times A \rightarrow A$  is a *combinatory algebra* if there are elements  $K, S \in A$  satisfying for all  $a, b, c \in A$

$$@(@(K, a), b) = a \quad (1)$$

$$@(@(@(S, a), b), c) = @(@(a, c), @(b, c)) \quad (2)$$

- (a) Show that there is a binary operation on the set of equivalence classes of closed  $\lambda$ -terms for the equivalence relation of  $\beta$ -conversion that makes it a combinatory algebra. [5 marks]

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*Answer:* Let  $[M]$  denote the equivalence class of a closed  $\lambda$ -term  $M$  under the equivalence relation of  $\beta$ -conversion,  $=_\beta$ .

Since  $=_\beta$  is a congruence for application (i.e.  $M =_\beta M'$  and  $N =_\beta N'$  implies  $MN =_\beta M'N'$ ), application of  $\lambda$ -terms induces a well-defined binary operation on equivalence classes satisfying  $@([M], [N]) = [MN]$ .

We define  $K = [\lambda xy. x]$  and  $S = [\lambda xyz. xz(yz)]$ . For all  $a, b, c$  there are  $\lambda$ -terms  $M, N, P$  with  $a = [M]$ ,  $b = [N]$  and  $c = [P]$  and hence

$$@(@(K, a), b) = [(\lambda xy. x) M N] = [M] = a$$

$$@(@(@(S, a), b), c) = [(\lambda xyz. xz(yz)) M N P] = [(M P)(N P)] = @(@(a, c), @(b, c))$$


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- (b) Show that every combinatory algebra  $A$  contains an element  $I$  satisfying

$$@ (I, a) = a \quad (3)$$

for all  $a \in A$ . [Hint: what does (2) tell us when  $a = b = K$ ?] [2 marks]

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*Answer:* Define  $I = @(@ (S, K), K)$ . Then

$$@ (I, a) = @ (@ (@ (S, K), K), a) \stackrel{(2)}{=} @ (@ (K, a), @ (K, a)) \stackrel{(1)}{=} a$$


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- (c) For an arbitrary combinatory algebra  $A$ , let  $A[x]$  denote the set of expressions given by the grammar

$$e ::= x \mid \ulcorner a \urcorner \mid (ee)$$

where  $x$  is some fixed symbol not in  $A$  and  $a$  ranges over the elements of  $A$ . Given  $e \in A[x]$  and  $a \in A$ , let  $e[x := a]$  denote the element of  $A$  resulting from interpreting occurrences of  $x$  in  $e$  by  $a$ , interpreting the expressions of the form  $\ulcorner a' \urcorner$  by  $a'$  and interpreting expressions of the form  $(ee')$  using  $@$ .

- (i) Give the clauses in a definition of  $e[x := a]$  by recursion on the structure of  $e$ . [2 marks]

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*Answer:*

$$x[x := a] = a \quad (4)$$

$$\ulcorner a' \urcorner[x := a] = a' \quad (5)$$

$$(ee')[x := a] = @ (e[x := a], e'[x := a]) \quad (6)$$


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- (ii) For each  $e \in A[x]$  show how to define an element  $\Lambda_x e \in A$  with the property that

$$@(\Lambda_x e, a) = e[x := a] \quad (7)$$

for all  $a \in A$ .

[6 marks]

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*Answer:* We can give the definition of  $\Lambda_x e$  by recursion on the structure of  $e$ :

$$\Lambda_x x = I \quad (8)$$

$$\Lambda_x \ulcorner a' \urcorner = @ (K, a) \quad (9)$$

$$\Lambda_x (ee') = @ (@ (S, \Lambda_x e), \Lambda_x e') \quad (10)$$

where  $I$  is as in part (b). For then we have

$$\begin{aligned} @(\Lambda_x x, a) &\stackrel{(8)}{=} @(I, a) \stackrel{(3)}{=} a \stackrel{(4)}{=} x[x := a] \\ @(\Lambda_x \ulcorner a' \urcorner, a) &\stackrel{(9)}{=} @(@ (K, a'), a) \stackrel{(1)}{=} a' \stackrel{(5)}{=} a'[x := a] \\ @(\Lambda_x (ee'), a) &\stackrel{(10)}{=} @(@ (@ (S, \Lambda_x e), \Lambda_x e'), a) \stackrel{(2)}{=} @(@ (\Lambda_x e, a), @(\Lambda_x e', a)) \\ &\stackrel{\text{rec. hyp.}}{=} @(e[x := a], e'[x := a]) \stackrel{(6)}{=} (ee')[x := a] \end{aligned}$$


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- (d) Recall the usual encoding of Booleans in  $\lambda$ -calculus. Using Part (c)(ii), show that in any combinatory algebra  $A$  there are elements  $True, False \in A$  and a function  $If : A \times A \rightarrow A$  satisfying

$$@ (If(a, b), True) = a \quad (11)$$

$$@ (If(a, b), False) = b \quad (12)$$

for all  $a, b \in A$

[5 marks]

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*Answer:* In the  $\lambda$ -calculus we can take  $True = \lambda xy.x$ ,  $False = \lambda xy.y$  and  $If(M, N) = \lambda x.xMN$ . So in a combinatory algebra we can take

$$True = K \quad (13)$$

$$False = @ (K, I) \quad (14)$$

$$If(a, b) = \Lambda_x ((x \ulcorner a \urcorner) \ulcorner b \urcorner) \quad (15)$$

For then we have:

$$@ (If(a, b), True) \stackrel{(15)+(7)}{=} @ (@ (True, a), b) \stackrel{(13)}{=} @ (@ (K, a), b) \stackrel{(1)}{=} a$$

and

$$@ (If(a, b), False) \stackrel{(15)+(7)}{=} @ (@ (False, a), b) \stackrel{(14)}{=} @ (@ (@ (K, I), a), b) \stackrel{(1)}{=} @ (I, b) \stackrel{(3)}{=} b$$

[Unfolding the definitions in (15), we have that  $If(a, b)$  is

$$@ (@ (S, @ (@ (S, I), @ (K, a))), @ (K, b)).]$$


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